English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, Groups 4/5, 2020/2021 Test 1

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Recall that, for $A \subset \Omega$, we write $\overline{A} = \Omega \setminus A$.

Problem 1. (1 point) Three dice are rolled together five times. What is the probability that the number of rolls showing only odd values is even? Find the expectation of this count.

Problem 2. (0.8 points) Two numbers are chosen at random from $[0, 1]$. What is the probability that the sum of their squares exceeds 1?

Problem 3. (1 point) A race has 25 teams: 8 in the jeep category, 10 in trucks and 7 in motorcycles. Jeeps finish with probability 0.9, trucks with probability 0.7, and motorcycles with probability 0.6. After the race, three teams are chosen randomly from the participants. It is known that one of the selected teams finished and the other two did not. What is the probability that the three selected teams belong to different categories?

Problem 4. (1.2 points)

1. (0.4 points) Let $A, B \subset \Omega$ be events. Recall the definition of independence of A and B. Prove that if A and B are independent, then $\overline{A}$ and B are independent too.

2. (0.4 points) Let $n > 1$ be an integer and $A_1, \ldots, A_n$ events. Recall that $A_1, \ldots, A_n$ are mutually independent if, for every $\{i_1, \ldots, i_k\} \subset \{1, \ldots, n\}$,

$$\mathbb{P}(A_{i_1})\mathbb{P}(A_{i_2}) \ldots \mathbb{P}(A_{i_k}) = \mathbb{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) \,. \tag{1}$$

   Prove that if $A_1, \ldots, A_n$ are mutually independent, then $\overline{A_1}, \ldots, \overline{A_n}$ are also mutually independent.

3. (0.4 points) Let X take values in $\mathbb{N}$, and for $n \in \mathbb{N}$ put $p_n := \mathbb{P}(X = n)$. Define the expectation $\mathbb{E}X$ and variance DX. Prove that

$$\sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \mathbb{E}X \,. \tag{2}$$